

PART A Series

$$a. R_1 + R_2 + R_3 \\ 20 + 30 + 50 = 100 \Omega$$

Total current

$$I = \frac{V}{R} = \frac{125}{100} \\ = 1.25 A$$

Current for each Resistor

$$1.25 A$$

Voltage drop for each

$$R_1 = V = I \times R$$

$$= 1.25 \times 20 = 25 V$$

$$R_2 = 1.25 \times 30 = 37.5 V$$

$$R_3 = 1.25 \times 50 = 62.5 V$$

Power through each Resistor

$$P = V \times I$$

$$R_1 = 25 \times 1.25 = 31.25 W$$

$$R_2 = 37.5 \times 1.25 = 46.875 W$$

$$R_3 = 62.5 \times 1.25 = 78.125 W$$

Parallel

$$\frac{1}{R_{Total}} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = \frac{8}{100}$$

$$R_{Total} = 12.5 \Omega$$

Current

$$I = \frac{125}{12.5} = 10 A$$

Current through each Resistor

$$I = \frac{V}{R} = \frac{125}{20} = 6.25 A$$

$$\frac{125}{100} = 1.25 A$$

$$\frac{125}{50} = 2.5 A$$

Voltage drop

$$125 V$$

Power

$$R_1 = 125 \times 6.25 = 781.25 W$$

$$R_2 = 125 \times 1.25 = 156.25 W$$

$$R_3 = 125 \times 2.5 = 312.5 W$$